

HEALTHCARE ANALYTICS DASHBOARD

Week 6 Task Report

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1. Introduction & Objective

This Week 6 task focuses on developing a complete Healthcare Analytics dashboard in Power BI using hospital admission and billing information. The work includes data modelling, DAX measure creation, visual planning, and dashboard preparation. The main objective is to provide a single view of patient count, billing performance, and average stay, with filtering based on insurance provider, admission type, and admission date. A readmission-rate calculation is also included for analysis.

1.1 Dataset Overview

The healthcare_dataset contains one record for each billing entry and includes the following information:

- Patient details: Name, Age, Age Group, Gender, Blood Type
- Admission details: Admission ID, Admission Type, Date of Admission, Discharge Date, Doctor, Hospital, Room Number
- Clinical details: Medical Condition, Medication, Test Results
- Financial details: Billing Amount, Insurance Provider

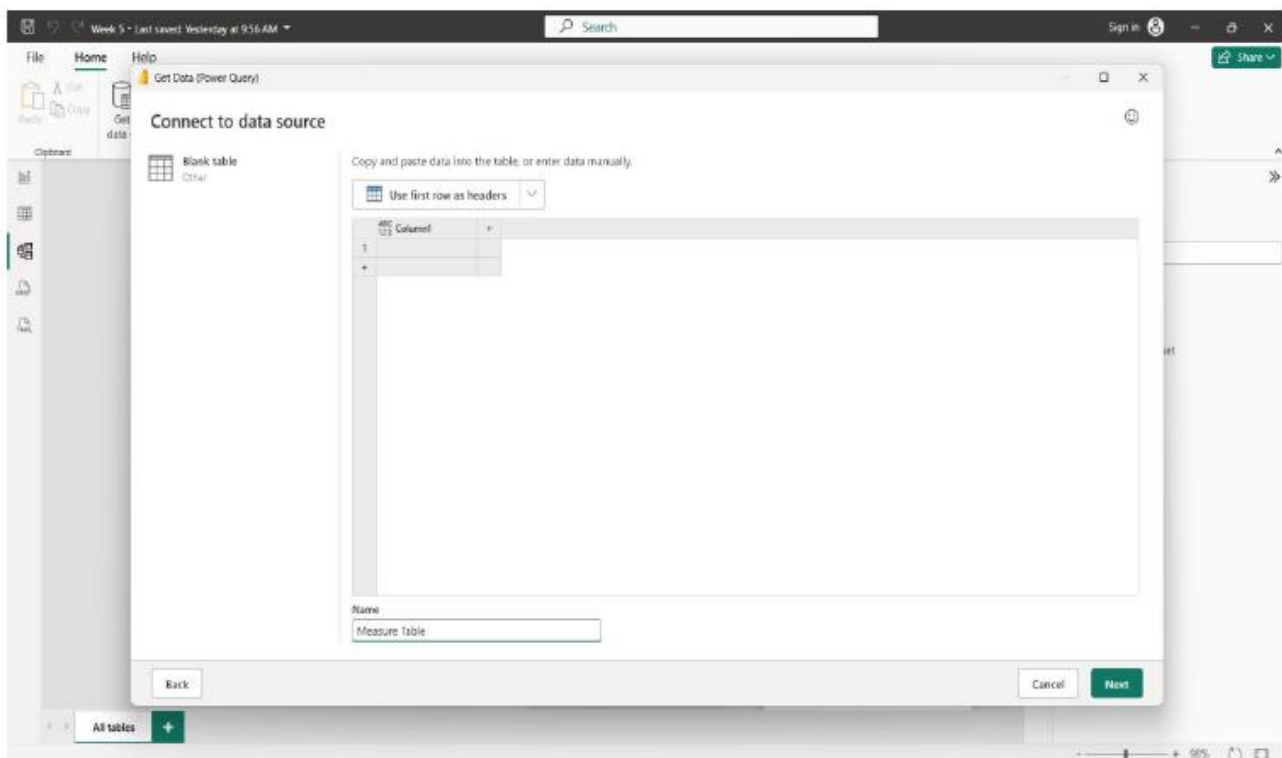
2. Data Model Setup

Before creating the DAX calculations, a separate Measure Table was prepared so that all calculated measures could be maintained independently from the source tables. The blank-table option in Power BI was used to create this dedicated location for the measures.

The final structure follows a star-schema arrangement. Billing acts as the central fact table and is connected to Dim_Patient and Dim_Admission. The original healthcare_dataset is retained for reference, while the Measure Table stores the calculated measures.

Table roles:

- Billing — Fact table containing Admission ID, Billing Amount, Date of Admission, Discharge Date, Insurance Provider and Patient ID.
- Dim_Patient — Dimension containing Age, Age Group, Blood Type, Gender, Name and Patient ID.
- Dim_Admission — Dimension containing Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number and Test Results.
- Measure Table — Stores the DAX calculations without regular data columns.
- healthcare_dataset — Original flat source table kept for checking and validation.



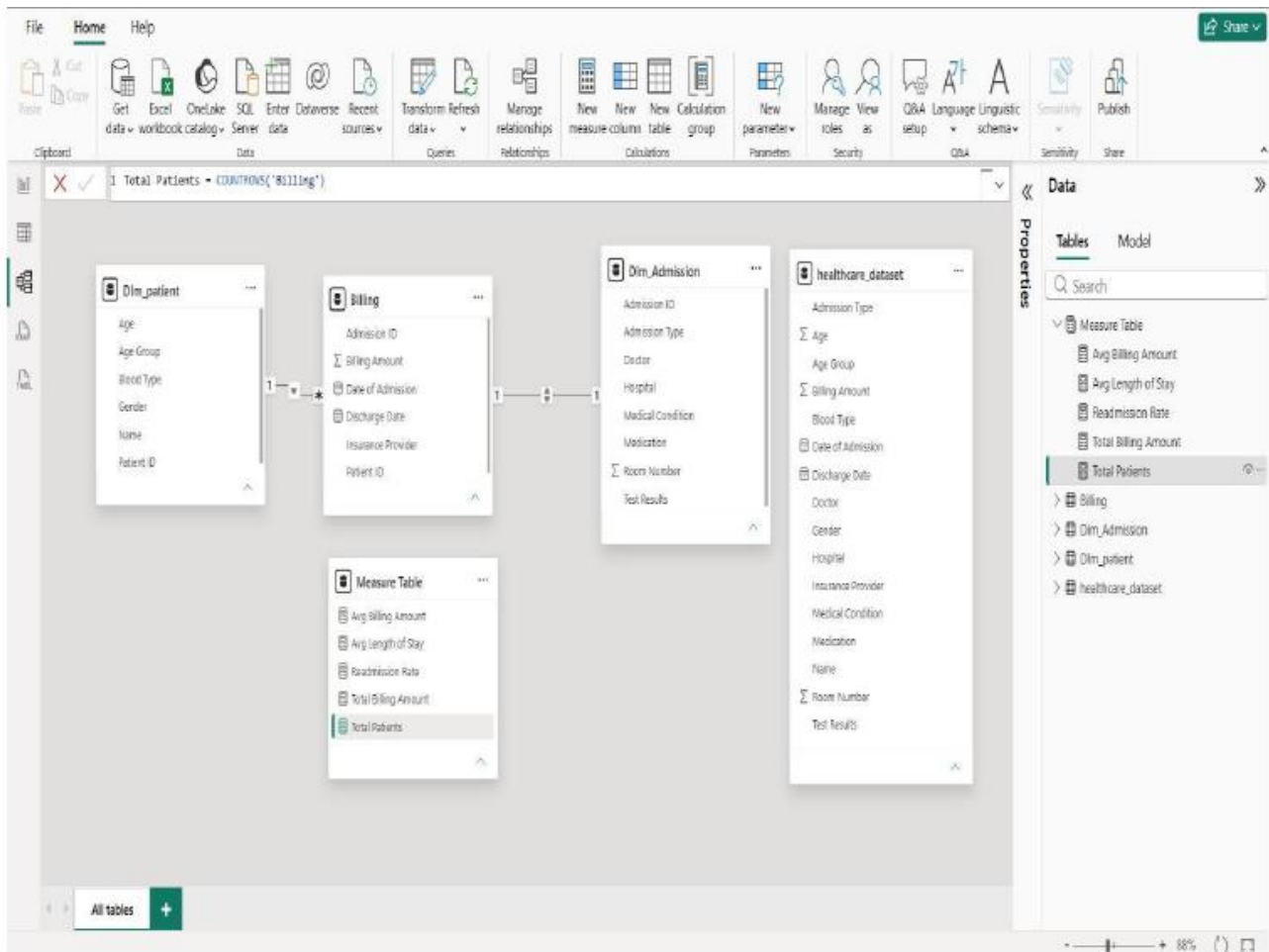
3. DAX Measures Development

Five important measures were created using DAX. Each calculation was added to the Measure Table separately and checked in the model before proceeding to the next measure.

3.1 Total Patients

This measure counts the records in the Billing table and provides the basic patient-volume indicator.

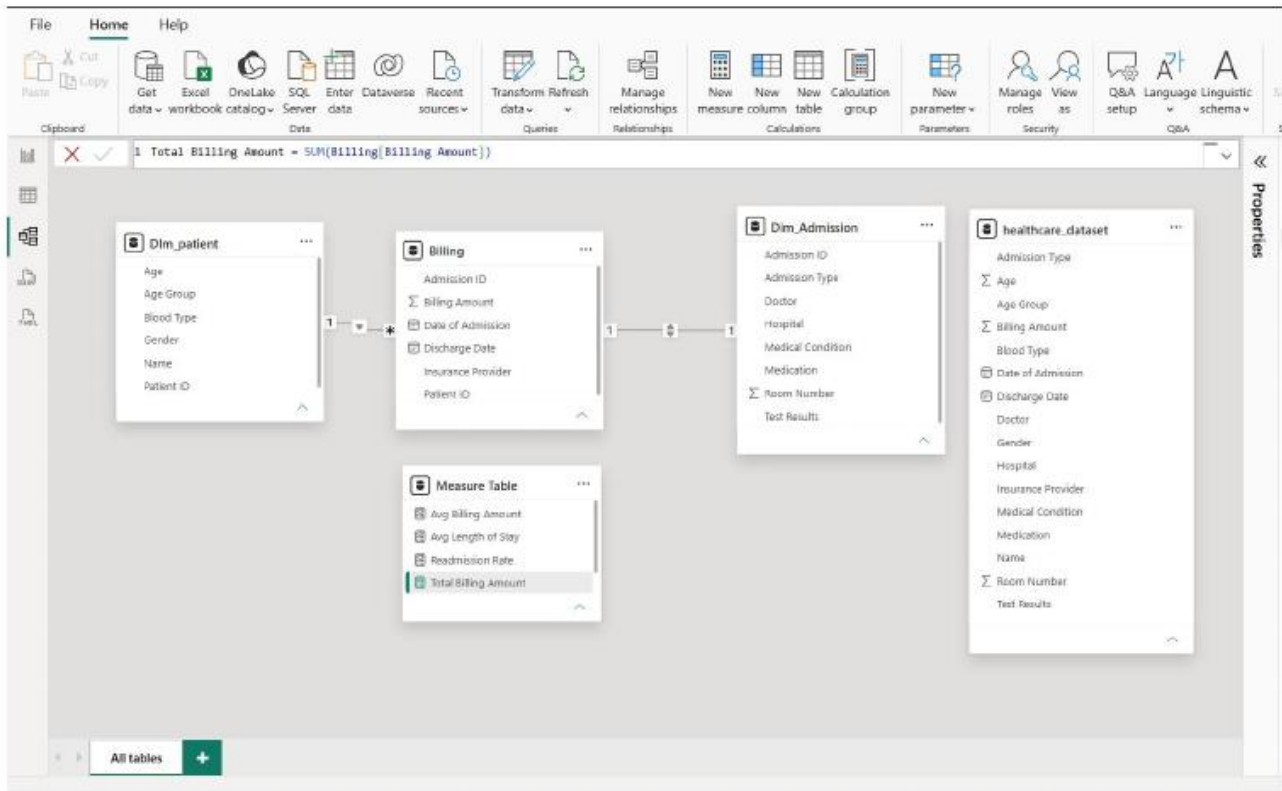
Total Patients = COUNTROWS(Billing)



3.2 Total Billing Amount

This calculation adds all billing values together to obtain the overall billing amount.

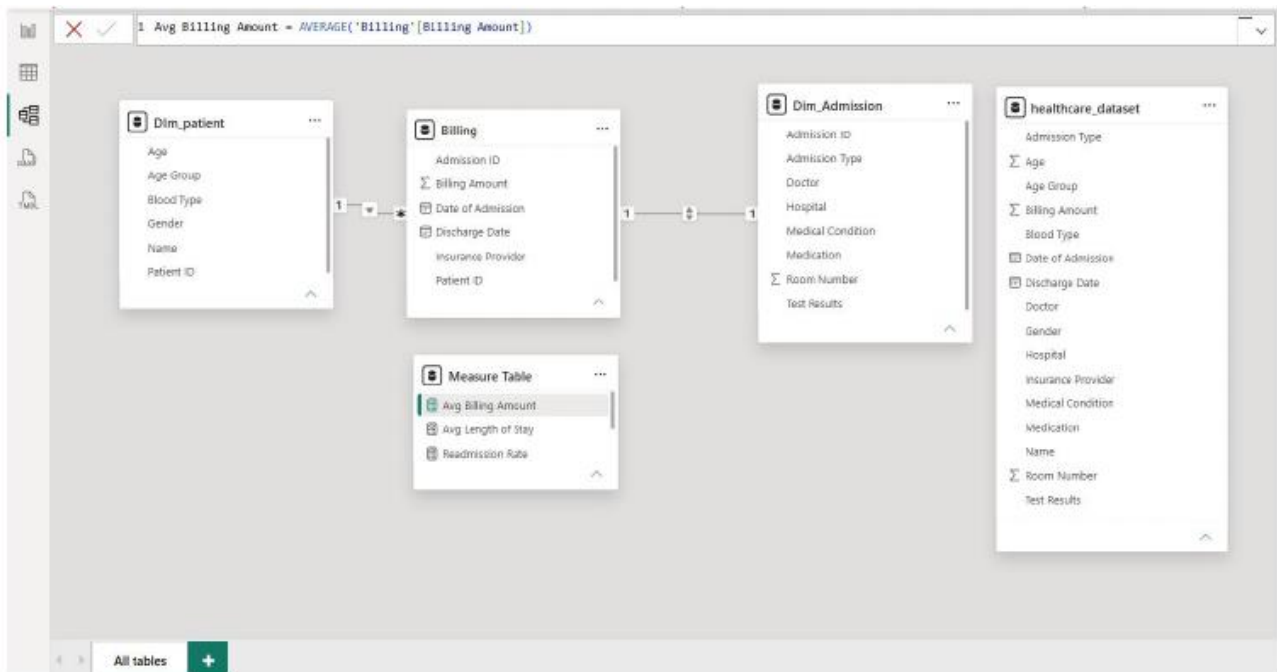
Total Billing Amount = SUM(Billing[Billing Amount])



3.3 Average Billing Amount per Patient

This measure calculates the mean billing value for the records and can be used to understand the typical cost associated with a billing entry.

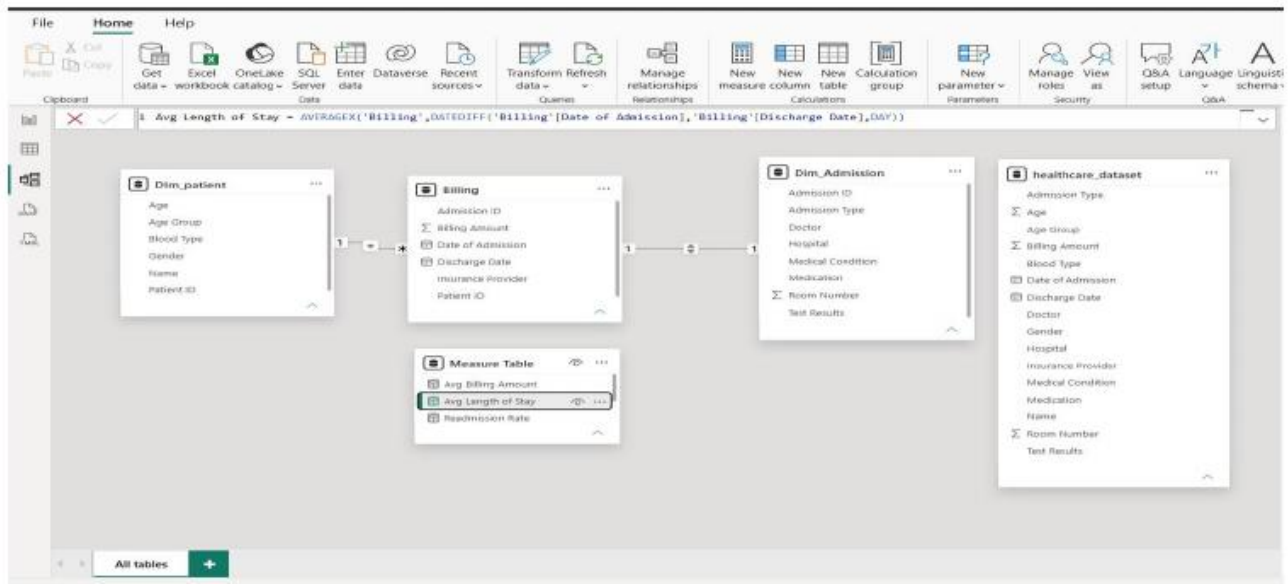
Avg Billing Amount = AVERAGE(Billing[Billing Amount])



3.4 Average Length of Stay (Days)

The average stay is obtained by finding the number of days between admission and discharge for each billing record and then averaging those values.

Avg Length of Stay = AVERAGEX(Billing, DATEDIFF(Billing[Date of Admission], Billing[Discharge Date], DAY))



3.5 Readmission Rate

The readmission calculation identifies patients with more than one admission in the Billing table. It groups records by Patient ID, filters patients whose admission count is greater than one, and then divides that count by the total patient count.

Readmission Rate = DIVIDE(RepeatPatients, TotalCount, 0)

The screenshot displays the Microsoft Power BI Desktop interface. At the top, the ribbon shows various tabs including File, Home, and Help. Below the ribbon, the DAX query editor contains the following code:

```
1 Readmission Rate = VAR TotalCount = [Total Patients]
2 VAR RepeatPatients = COUNTROWS(FILTER(SUMMARIZE('Billing','Billing'[Patient ID],'Admission Count',COUNTROWS('Billing')),[Admission Count] > 1))
3 RETURN DIVIDE(RepeatPatients,TotalCount,0)
```

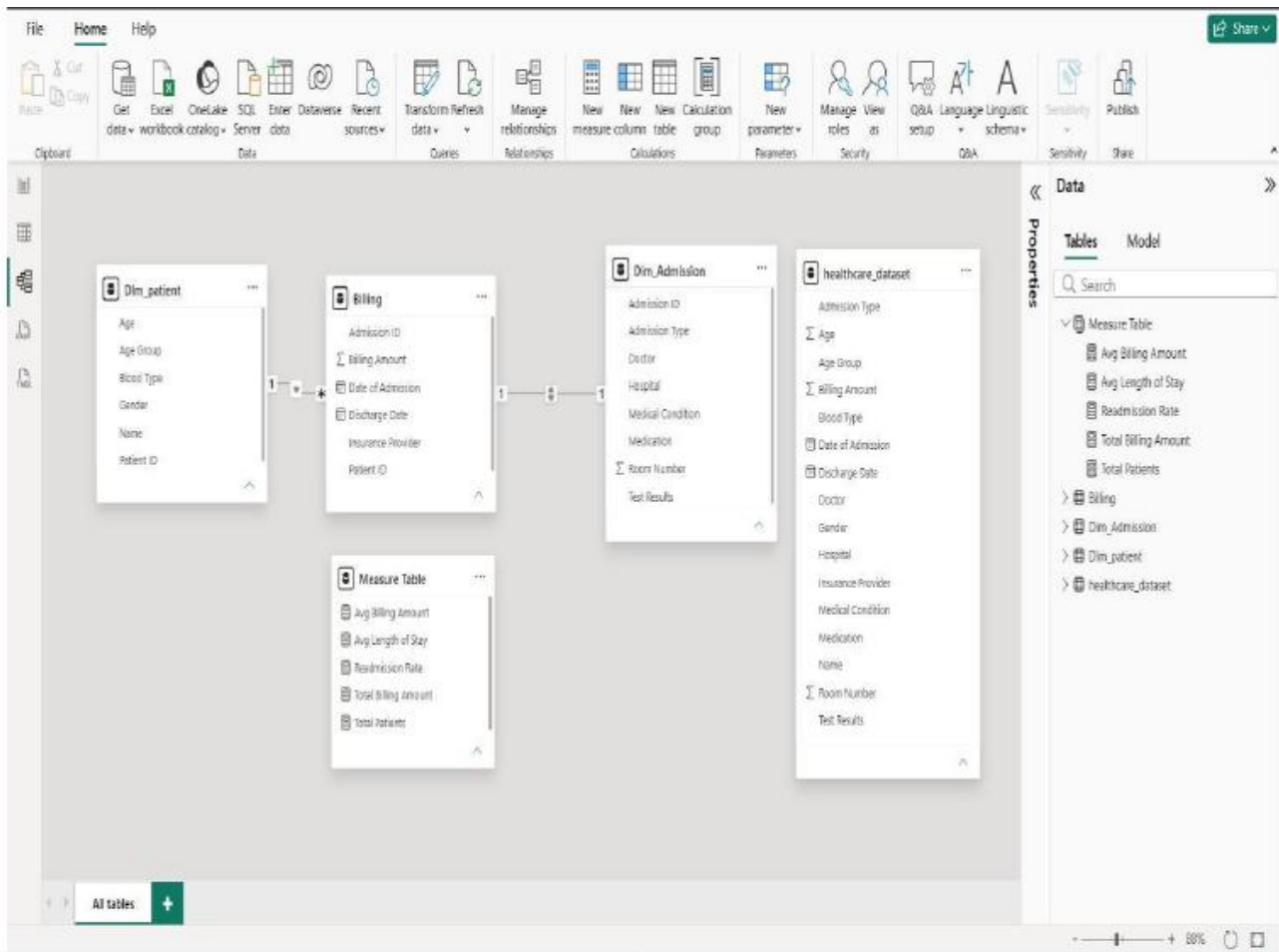
The main workspace shows a data model diagram with four tables: Dim_patient, Billing, Dim_Admission, and healthcare_dataset. The Dim_patient table is connected to the Billing table, which is connected to the Dim_Admission table. The healthcare_dataset table is also connected to the Dim_Admission table. A Measure Table is shown in the center, containing the following measures:

- Avg Billing Amount
- Avg Length of Stay
- Readmission Rate
- Total Billing Amount
- Total Patients

The Readmission Rate measure is highlighted. On the right side, the Properties pane shows the Data tab, which lists the tables and measures used in the model.

3.6 All Measures

After completing the five calculations, the Measure Table contains Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount and Total Patients.



4. Visual Layout & Dashboard Design

After completing the measures, the dashboard structure was planned so that every visual has a specific purpose and a clear connection to the available fields. The layout was organized to present summary information first and detailed analysis below it.

4.1 Top KPI Summary Cards

- Total Patients
- Total Billing Amount
- Average Billing Amount
- Average Length of Stay

These four cards provide a quick summary of the main healthcare and billing indicators.

4.2 Demographics & Patient Distribution

Matrix: Age Group is placed in Rows, Medical Condition in Columns, with Total Patients and Average Billing Amount as values.

Donut chart: Age Group is used as the legend and Total Patients as the value to compare patient distribution across age categories.

4.3 Medical Conditions & Admission Volumes

A stacked bar chart compares patient counts by Medical Condition, with Admission Type used to separate the admission categories.

4.4 Admission Volume Trends & Hospital Analytics

A line chart displays admission and billing trends using Date of Admission by Year and Month. A column-based visual compares Total Billing Amount across Insurance Providers.

4.5 Slicers & Dynamic Cross-Filtering

- Insurance Provider slicer
- Admission Type slicer
- Date of Admission range slicer

These controls allow the dashboard to be filtered dynamically according to the selected provider, admission category, or time period.

5. Report Building Process

The report visuals were arranged on the Power BI canvas in stages. Placeholder visuals were placed first, followed by assigning the required fields and measures to each visual.

5.1 Adding KPI Card Placeholders

A Card visual was inserted and kept as an empty placeholder initially. It was then duplicated to create four KPI spaces.

5.2 Adding the Matrix Visual

A Matrix visual was positioned beside the cards and prepared with Age Group, Medical Condition, Total Patients and Average Billing Amount.

5.3 Adding the Donut Chart

A Donut chart was added to represent the patient distribution by Age Group.

5.4 Adding the Stacked Bar Chart

A stacked bar chart was created to compare patients by Medical Condition and Admission Type.

5.5 Adding the Line Chart

A line chart was placed for analysing admission and billing changes over time.

5.6 Adding the Line and Clustered Column Chart

A Line and clustered column visual was placed for the Insurance Provider billing comparison, using the column part for the displayed values.

5.7 Adding the Slicers

Three slicers were added for Insurance Provider, Admission Type and Date of Admission.

5.8 Binding Data to the KPI Cards

The four planned measures were assigned to the KPI cards: Total Patients, Total Billing Amount, Avg Billing Amount and Avg Length of Stay.

5.9 Building the Trend Line Chart

Date of Admission using Month and Year was placed on the X-axis, while Total Patients and Total Billing Amount were used for the trend values.

5.10 Building the Billing by Insurance Provider Chart

Insurance Provider was placed on the X-axis and Total Billing Amount was assigned as the column value.

5.11 Configuring the Insurance Provider Slicer

The Insurance Provider field was added to the first slicer so the report could be filtered by provider. The Admission Type slicer was configured similarly.

5.12 Configuring the Date of Admission Slicer

Date of Admission was assigned to the final slicer, creating a date-range control for filtering the report.

OUTPUT